

Hospital Appointment Booking System (Cloud-Based Healthcare Management Solution using AWS)

Prepared in the partial fulfillment of the Summer Internship Program on AWS
AT



Under the guidance of
Mrs. Sumana Bethala, APSSDC
Mr. Juttuka Lovababu, APSSDC

Submitted by

KOTTAPALLI INDIRA PRIYA DARSHINI	– 23001A0522	JNTUACEA ANANTAPUR, (Batch -1)
N SRAVANI	– 23001A0527	JNTUACEA ANANTAPUR, (Batch -1)
BANGARU RAJAVARDHAN	– 23001A0505	JNTUACEA ANANTAPUR, (Batch -1)
SHAIK MAHAMAD APSAR	– 23001A0546	JNTUACEA ANANTAPUR, (Batch -1)
VASIPALLI CHETHAN REDDY	– 23001A0541	JNTUACEA ANANTAPUR, (Batch -1)

Contents

1. INTRODUCTION

1.1 Introduction to the Project.....	6
1.2 Need for the Project.....	6
1.3 Objectives of the Project.....	7
1.4 Scope of the Project	7

2. METHODOLOGY

2.1 Methodology Overview	8
2.2 Requirement Analysis	9
2.3 System Development Approach.....	10
Frontend Development	10
Backend Development	10
Database & AWS Integration	10
2.4 Appointment Booking Process	10
2.5 Notification Methodology.....	11
2.6 Testing Methodology	12

3 .TECHNOLOGY STACK

3.1 Frontend Technologies	13
3.2 Backend Technologies	13
3.3 Database	14
3.4 AWS Cloud Services.....	14

4. SYSTEM DESIGN AND ARCHITECTURE

4.1 Overall Architecture	15
4.2 Frontend Architecture	16
4.3 Backend Architecture	16
4.4 Data Flow Processes.....	17
4.5 Database Design	18
4.6 Security Design	19

5. IMPLEMENTATION

5.1 Frontend Implementation.....	21
5.2 Backend Implementation.....	21
5.3 Authentication Implementation.....	22
5.4 Appointment Booking.....	22
5.5 Admin Module	22
5.6 Notification.....	22
5.7 Deployment.....	23

6. RESULTS

6.1 Final Output	24
6.2 Functional Results	24
6.3 AWS Integration Results	24
6.4 Deployment Results	25
6.5 Testing Results	25
6.6 Output Screenshots	26

7. CONCLUSION

<u>Conclusion</u>	26
-------------------------	----

Abstract

The Hospital Appointment Booking System using AWS is a cloud-based web application developed to simplify and modernize the process of scheduling medical appointments. Traditional appointment systems often rely on manual record management, phone calls, or in-person visits, leading to long waiting times, scheduling conflicts, inefficient resource utilization, and limited accessibility for patients. This project addresses these challenges by providing a secure, user-friendly, and scalable online platform that enables patients to book appointments with doctors from anywhere at any time while allowing healthcare administrators to efficiently manage appointments, doctors, and patient records.

The application is built using a modern full-stack architecture. The frontend is developed using **React.js**, providing a responsive and interactive user interface for patients, doctors, and administrators. The backend is implemented using **Node.js** and **Express.js**, which expose RESTful APIs to handle authentication, appointment scheduling, doctor management, and administrative operations. **MongoDB Atlas** serves as the primary cloud database for storing user information, doctor details, appointments, and application data. To enhance scalability and reliability, the project integrates multiple **Amazon Web Services (AWS)** components, including **Amazon S3** for hosting the frontend application, **Amazon SNS** for sending appointment confirmations and notifications, and **AWS IAM** for secure access management and resource permissions.

The system supports role-based access for three types of users: patients, doctors, and administrators. Patients can register, log in securely, browse available doctors, select suitable appointment slots, and manage their bookings. Doctors can view their appointment schedules, while administrators are responsible for managing doctor profiles, monitoring appointments, and maintaining the overall functionality of the system. Secure authentication, efficient database management, and cloud-based deployment ensure high availability, improved performance, and enhanced data security.

The successful implementation of this project demonstrates how cloud computing technologies can improve healthcare service delivery by automating appointment scheduling and reducing administrative overhead. The proposed system offers an efficient, scalable, and reliable solution that enhances the patient experience while simplifying hospital management. Furthermore, the application provides a strong foundation for future enhancements such as online payment integration, telemedicine support, AI-based appointment recommendations, electronic medical records, and mobile application development, making it suitable for deployment in modern healthcare environments.

Acknowledgement

I would like to express my heartfelt gratitude to all those who have contributed to the successful completion of my summer internship project at Andhra Pradesh Skill Development Corporation (APSSDC). This opportunity has been an enriching and transformative experience for me, and I am truly thankful for the support, guidance, and encouragement I have received along the way.

First and foremost, I extend my sincere regards to Mrs Sumana Bethala and Mr Juttuka Lovababu, my supervisor and mentor, for providing me with valuable insights, constant guidance, and unwavering support throughout the duration of the internship. Her expertise and encouragement have been instrumental in shaping the direction of this project.

I would like to thank the entire team at Andhra Pradesh Skill Development Corporation (APSSDC) for fostering a collaborative and innovative environment. The camaraderie, knowledge sharing, and feedback I received from my colleagues significantly contributed to the development and success of this project.

In conclusion, I am honoured to have been a part of this internship program, and I look forward to leveraging the skills and knowledge gained to contribute positively to future endeavours.

Thank you.

Sincerely,

KOTTHAPALLI INDHIRA PRIYA DHARSHINI	-- 23001A0522N
SRAVANI	--23001a0527SHAIK
MAHAMAD APSAR	--23001a0546BANGARU
RAJAVARDHAN	-- 23001a0505VASIPALLI CHETHAN
REDDY	--23001a0541

Chapter 1

INTRODUCTION

1.1 Introduction to the Project

The Hospital Appointment Booking System using Amazon Web Services (AWS) is a cloud-based healthcare application developed to simplify the appointment scheduling process and improve the efficiency of hospital management. In many hospitals, appointment booking is still performed manually through phone calls or direct visits, which often leads to long waiting times, scheduling conflicts, increased administrative workload, and poor patient experience. With the growing demand for digital healthcare services, there is a need for an automated, secure, and easily accessible appointment management system.

This project provides an online platform where patients can register, search for available doctors, and book appointments based on their preferred date and time. Doctors can review appointment requests, manage their schedules, and update consultation statuses, while administrators can monitor doctors, patients, and overall system activities through a centralized dashboard.

To ensure scalability, security, and high availability, the system is deployed using **Amazon Web Services (AWS)**. The application utilizes **Amazon S3** for hosting the frontend, **Amazon SNS** for sending appointment notifications, **AWS IAM** for secure access management, and **MongoDB Atlas** as the cloud database for storing application data. These cloud technologies provide reliable performance, secure data management, and efficient communication between users.

The proposed system offers a modern and user-friendly solution that enhances healthcare service delivery by reducing manual effort, minimizing appointment delays, improving communication, and ensuring better accessibility for both patients and healthcare providers.

1.2 Need for the Project

The increasing demand for healthcare services requires a faster and more efficient appointment management system. Traditional appointment booking methods, such as phone calls and manual registration, often result in long waiting times, scheduling conflicts, and increased administrative workload. Patients also face difficulties in checking doctor availability and receiving timely appointment updates.

The **Hospital Appointment Booking System** addresses these challenges by providing a secure cloud-based platform where patients can book appointments online, doctors can

efficiently manage schedules, and administrators can monitor hospital operations. By integrating **AWS cloud services**, the system ensures scalability, data security, reliable storage, and automated notifications, thereby improving the overall healthcare experience and operational efficiency.

1.3 Objectives of the Project

The primary objective of the **Hospital Appointment Booking System** is to develop a secure and cloud-based platform that simplifies the appointment scheduling process for patients, doctors, and hospital administrators. The system aims to reduce manual paperwork, minimize waiting times, and improve the overall efficiency of hospital operations. It enables patients to register, log in, search for doctors, and book appointments online, while allowing doctors to manage appointment requests effectively. The project also focuses on providing centralized administration, secure data management, and automated appointment notifications using AWS cloud services. Additionally, it aims to deliver a scalable, reliable, and user-friendly healthcare solution that enhances communication and accessibility for all users

1.4 Scope of the Project

The scope of the Hospital Appointment Booking System includes developing a cloud-based application that enables efficient appointment scheduling and hospital management. The system provides separate functionalities for patients, doctors, and administrators. Patients can register, log in, search for doctors, and book appointments online. Doctors can view and manage appointment requests, while administrators can monitor doctors, patients, and appointment records through a centralized dashboard. The project also includes secure data management, role-based access, and automated email notifications using AWS services. Furthermore, the system is designed to be scalable and can be enhanced with features such as online payments, telemedicine, mobile applications, and AI-based healthcare recommendations in the future.

Chapter 2

METHODOLOGY

2.1 Methodology Overview

The development of the Hospital Appointment Booking System followed a systematic and modular methodology to ensure efficient design, implementation, and deployment. The project was divided into multiple phases, including requirement analysis, system design, frontend and backend development, database integration, AWS cloud service implementation, testing, and deployment. Each module was developed and tested independently before being integrated into the complete system. This approach helped identify and resolve issues at an early stage while ensuring better code maintainability and system reliability.

The methodology adopted for this project consists of the following phases:

1. **Requirement Analysis** – Identifying user requirements and system functionalities.
2. **System Design** – Designing the application architecture, database, and user interfaces.
3. **Frontend Development** – Building an interactive user interface using React.js.
4. **Backend Development** – Developing RESTful APIs using Node.js and Express.js.
5. **Database Integration** – Storing and managing application data using MongoDB Atlas.
6. **AWS Cloud Integration** – Implementing Amazon S3, SNS, and IAM for cloud deployment and security.
7. **Testing and Validation** – Performing functional and integration testing to ensure system reliability.
8. **Deployment** – Deploying the application on the cloud and verifying its performance.

This structured methodology ensured the successful development of a secure, scalable, and user-friendly hospital appointment management system.

2.2 Requirement Analysis

Requirement analysis is a crucial phase in the development of the **Hospital Appointment Booking System**, as it identifies the functional and non-functional requirements needed to build an efficient and reliable application. During this phase, the needs of patients, doctors, and administrators were carefully analyzed to ensure that the system addresses real-world hospital appointment management challenges.

Functional Requirements

The system should provide the following functionalities:

- User registration and secure login for patients, doctors, and administrators.
- Search and view doctor information based on specialization.
- Book, view, and manage appointments online.
- Doctors can approve or reject appointment requests.
- Administrators can manage doctors, patients, and appointment records.
- Send appointment confirmation and status notifications using Amazon SNS.

Non-Functional Requirements

The system should satisfy the following quality requirements:

- **Security:** Protect user data through authentication and AWS IAM access control.
- **Scalability:** Support increasing numbers of users and appointments without affecting performance.
- **Availability:** Ensure the application remains accessible through cloud deployment.
- **Performance:** Process appointment requests quickly with minimal response time.
- **Usability:** Provide a simple, responsive, and user-friendly interface for all users.
- **Reliability:** Maintain consistent system performance and secure data management.

By identifying these requirements, the project was designed to provide a secure, scalable, and efficient hospital appointment management solution that meets the expectations of both healthcare providers and patients.

2.3 System Development Approach

The **Hospital Appointment Booking System** was developed using a **full-stack web development approach**, integrating modern web technologies with cloud computing services to build a secure, scalable, and user-friendly application. The development process was carried out in modular phases, allowing each component to be designed, implemented, tested, and integrated independently. This approach improved code maintainability, simplified debugging, and ensured seamless communication between the frontend, backend, database, and AWS services.

Frontend Development

The frontend was developed using **React.js**, providing a responsive and interactive user interface. It includes separate pages for user registration, login, patient dashboard, doctor dashboard, admin dashboard, appointment booking, and appointment management. React Router was used for navigation, while Axios was used to communicate with backend REST APIs.

Backend Development

The backend was implemented using **Node.js** and **Express.js**. It exposes RESTful APIs to handle user authentication, appointment scheduling, doctor management, and administrative operations. Business logic and request processing are managed through modular controllers and middleware.

Database & AWS Integration

MongoDB Atlas was used as the cloud database to store user details, doctor information, and appointment records. AWS services such as **Amazon S3** for frontend hosting, **Amazon SNS** for email notifications, and **AWS IAM** for secure access control were integrated to provide a scalable and reliable cloud-based solution.

This development approach ensured efficient implementation, secure data management, and smooth integration between all system components.

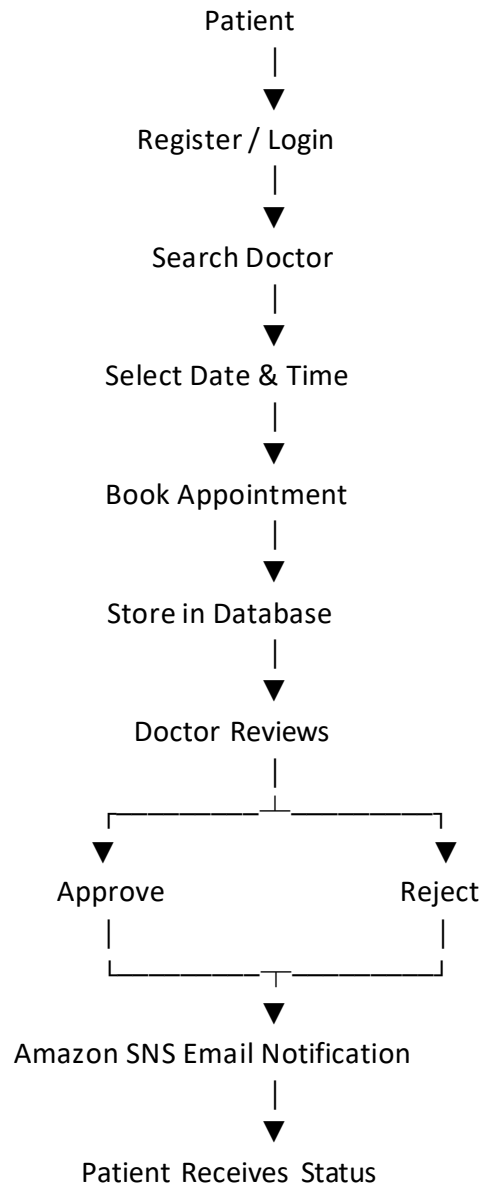
2.4 Appointment Booking Process

The appointment booking process is one of the core functionalities of the Hospital Appointment Booking System. It is designed to provide a simple, secure, and efficient method for scheduling medical appointments. The process begins when a patient registers and logs into the application using valid credentials. After successful authentication, the patient can browse the list of available doctors, view their specialization, and select a suitable doctor based on their healthcare requirements.

Once a doctor is selected, the patient chooses the preferred appointment date and time before submitting the booking request. The backend validates the request and stores the appointment details securely in the database. The appointment is then displayed on the doctor's dashboard, where the doctor can review and either approve or reject the request based on availability. After the appointment status is updated, the system automatically sends an email notification to the patient using **Amazon SNS**, informing them about the confirmation or rejection of the appointment.

This automated workflow eliminates manual scheduling, reduces appointment conflicts, improves communication between patients and doctors, and ensures efficient management of hospital appointments through a secure cloud-based platform.

The booking process is:



2.5 Notification Methodology

The **Hospital Appointment Booking System** uses **Amazon Simple Notification Service (Amazon SNS)** to send automated email notifications to patients regarding their appointment status. When a patient books an appointment, the request is stored in the database and forwarded to the respective doctor for review. Once the doctor approves or rejects the appointment, the backend triggers Amazon SNS to send an email notification to the patient's registered email address. This automated notification process improves communication, reduces manual effort, and ensures that patients receive timely updates about their appointments. The integration of Amazon SNS provides a reliable, scalable, and efficient cloud-based notification services.

2.6 Testing Methodology

The project was tested module by module to ensure the correctness, reliability, and smooth functioning of the application.

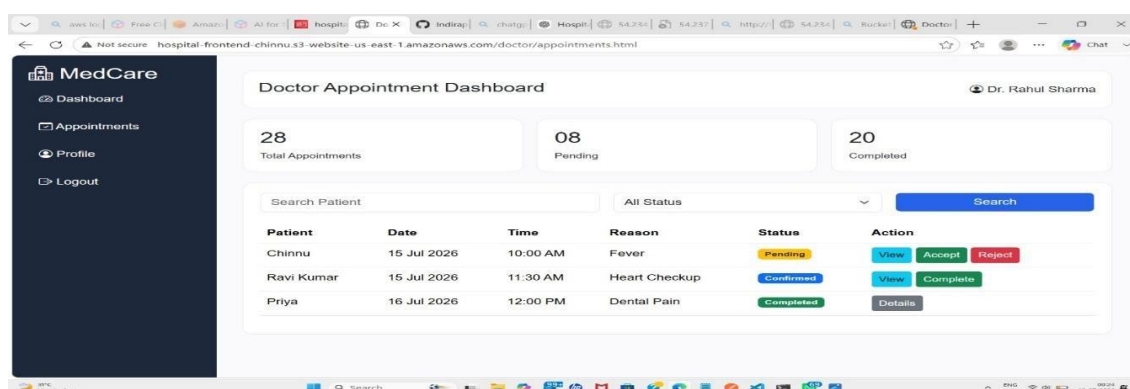
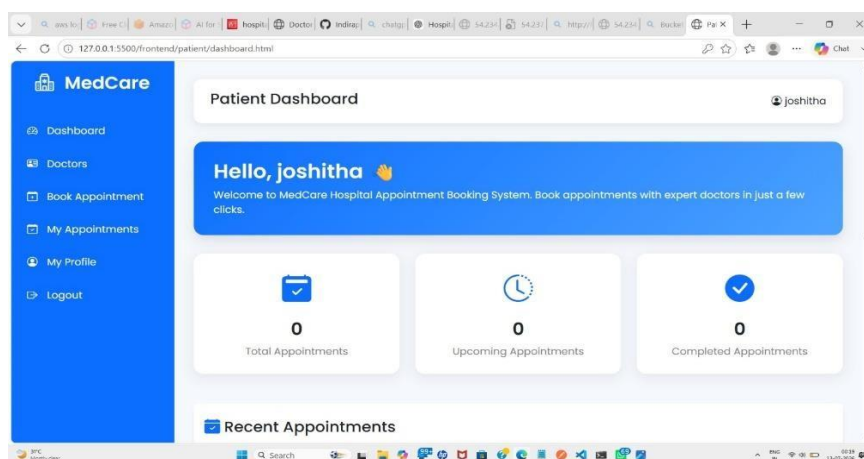
Authentication Testing: Verified patient, doctor, and administrator registration, secure login/logout functionality, and role-based access control. User authentication and session management were tested to ensure secure access to the system.

Appointment Testing: Validated appointment booking, viewing, approval, rejection, and status updates. The system was tested to ensure that appointment details were correctly stored and retrieved from the database.

Dashboard Testing: Verified the functionality of the Patient, Doctor, and Administrator dashboards. User-specific features such as appointment management, doctor management, and patient management were tested for accuracy and usability.

Notification Testing: Confirmed that Amazon SNS successfully sent email notifications whenever an appointment was approved or rejected. The notification content and delivery were verified using registered email accounts.

Deployment Testing: Ensured that the frontend hosted on Amazon S3 and the backend services functioned correctly after deployment. Frontend-backend communication, database connectivity, and AWS service integration were thoroughly tested to confirm reliable system performance.



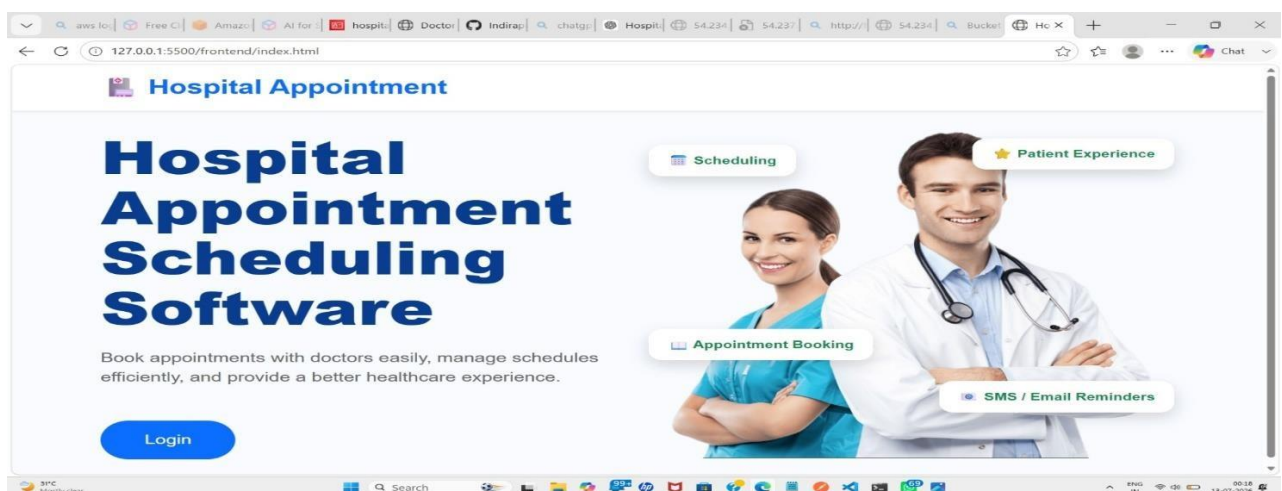
Chapter 3

TECHNOLOGY STACK

3.1 Frontend Technologies

The frontend of the Hospital Appointment Booking System was developed using the following technologies:

- **React.js:** Used to build a responsive and interactive user interface.
- **HTML5:** Provides the structure of web pages.
- **CSS3:** Used for styling and improving the visual appearance of the application.
- **JavaScript (ES6):** Implements dynamic functionality and client-side logic.
- **Axios:** Enables communication between the frontend and backend through REST APIs.
- **React Router DOM:** Handles navigation between different pages within the application.



3.2 Backend Technologies

The backend was developed using modern server-side technologies to handle application logic and API requests.

- **Node.js:** JavaScript runtime used for backend development.
- **Express.js:** Framework used to build RESTful APIs.
- **REST APIs:** Facilitate communication between the frontend and backend.
- **JWT Authentication:** Secures user authentication and authorization.
- **bcrypt:** Encrypts user passwords before storing them in the database.
- **dotenv:** Manages environment variables securely.

3.3 Database

The application uses a cloud database to store and manage hospital-related information.

- **MongoDB Atlas:** Stores patient, doctor, administrator, and appointment data.
- **Collections:** Maintains separate collections for users, doctors, appointments, and administrators.
- **Cloud Storage:** Provides secure, scalable, and reliable data management.
- **CRUD Operations:** Supports Create, Read, Update, and Delete operations efficiently.
- **Data Security:** Ensures secure storage and retrieval of application data.

3.4 AWS Cloud Services

The application integrates Amazon Web Services (AWS) to provide cloud-based hosting, notifications, and secure access management.

- **Amazon S3:** Hosts the frontend application using Static Website Hosting.
- **Amazon SNS:** Sends automated email notifications for appointment updates.
- **AWS IAM:** Provides secure authentication and permission management for AWS resources.
- **Cloud-Based Deployment:** Improves scalability, availability, and reliability of the application.
- **Secure Integration:** Ensures secure communication between the application and AWS services.

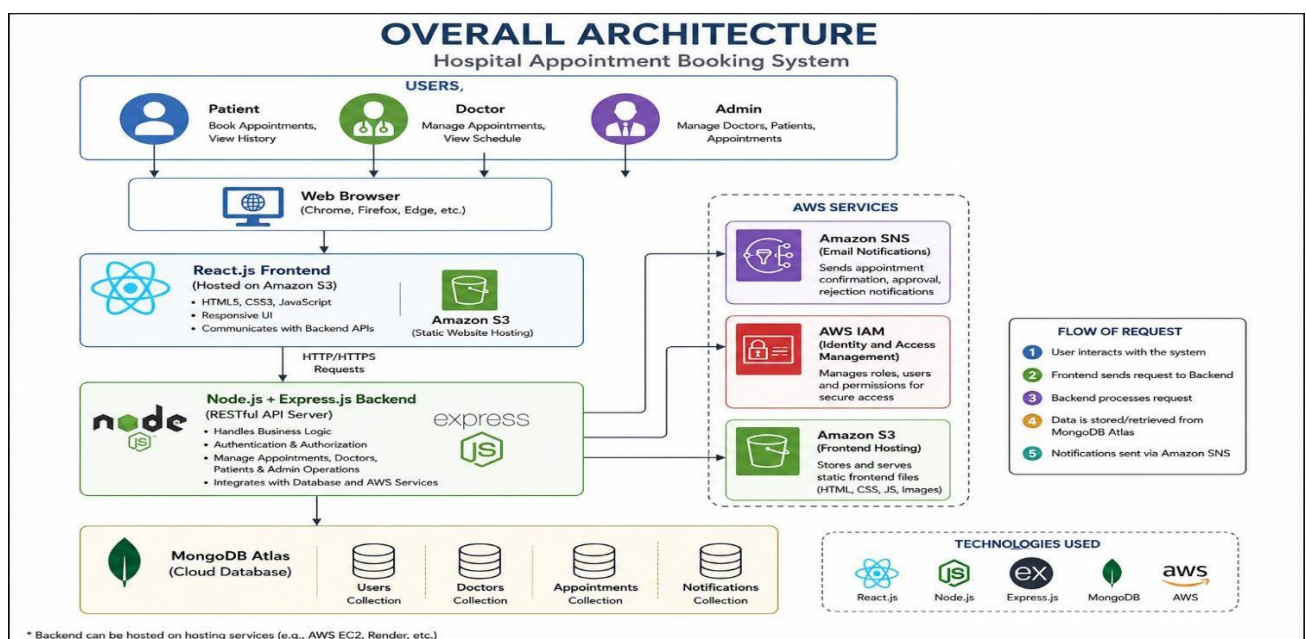
Chapter 4

SYSTEM DESIGN AND ARCHITECTURE

4.1 Overall Architecture

The Hospital Appointment Booking System follows a cloud-based client-server architecture that enables secure and efficient communication between patients, doctors, administrators, and cloud services. Users interact with the application through a web browser, where the frontend provides an intuitive interface for booking and managing appointments. The frontend communicates with the backend through RESTful APIs, which process user requests and perform business logic.

The backend interacts with the **MongoDB Atlas** database to store and retrieve user information, doctor details, and appointment records. AWS services such as **Amazon S3**, **Amazon SNS**, and **AWS IAM** are integrated to provide frontend hosting, email notifications, and secure access management. This architecture ensures scalability, security, high availability, and efficient data management while supporting multiple users simultaneously.



4.2 Frontend Architecture

The frontend of the Hospital Appointment Booking System is developed using React.js and follows a modular, component-based architecture to ensure code reusability, maintainability, and scalability. The application is organized into separate directories, each responsible for a specific functionality.

- **API Layer (api/):** Contains Axios configuration and API service files responsible for communicating with backend REST APIs for user authentication, doctor management, appointment booking, and administrative operations.
- **Components (components/):** Includes reusable UI components such as Navbar, Sidebar, Appointment Card, Doctor Card, Forms, Buttons, and Modal dialogs used across different pages.
- **Pages (pages/):** Contains complete application screens including Login, Registration, Patient Dashboard, Doctor Dashboard, Admin Dashboard, Appointment Booking, Doctor Management, Patient Profile, and Appointment History.
- **Routing (routes/):** Uses **React Router DOM** to enable seamless navigation between different pages based on user roles and authentication status.
- **Context & State Management:** Implements React Context API to manage user authentication, session information, role-based access, and shared application state across components.
- **Assets & Styling (assets/):** Stores application images, icons, logos, and CSS files to provide a responsive and visually appealing user interface.

4.3 Backend Architecture

The backend of the Hospital Appointment Booking System is developed using Node.js and Express.js, following a modular architecture that separates application logic, API routes, database operations, and cloud service integration. This structure improves maintainability, scalability, and efficient request handling.

- **Routes (routes/):** Defines RESTful API endpoints for user authentication, doctor management, patient management, appointment booking, and administrator operations.
- **Controllers (controllers/):** Contains the business logic for processing requests, validating user input, managing appointments, handling doctor approvals, and returning appropriate responses.
- **Models (models/):** Defines MongoDB schemas for Patients, Doctors, Administrators, and Appointments using Mongoose to ensure structured and consistent data storage.
- **Middleware (middleware/):** Implements authentication, authorization, request

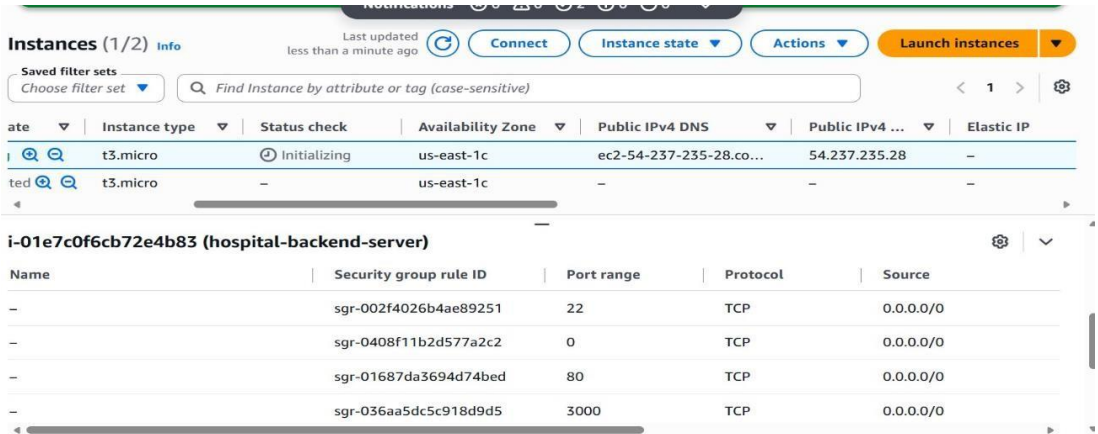
validation, and error-handling mechanisms to secure the application and protect API endpoints.

- **Database Layer:** Connects the backend with **MongoDB Atlas** to perform Create, Read, Update, and Delete (CRUD) operations efficiently.
- **AWS Integration:** Integrates **Amazon SNS** for sending appointment status notifications, **Amazon S3** for frontend hosting support, and **AWS IAM** for secure access to AWS resources.
- **Configuration (config/):** Stores database connection settings, environment variables, and AWS service configurations using **dotenv** for secure application deployment.

4.4 Data Flow Processes

The **Hospital Appointment Booking System** follows a structured data flow to ensure efficient communication between users, the backend, the database, and AWS services. User requests are processed through REST APIs, validated by the backend, and securely stored in the database. The system also integrates AWS services to provide notifications and cloud-based functionality.

- **Patient** registers or logs into the application using secure authentication.
- **Frontend (React.js)** collects user inputs and sends requests to the backend using REST APIs.
- **Backend (Node.js & Express.js)** validates the request and executes the required business logic.
- **MongoDB Atlas** stores and retrieves patient, doctor, and appointment information.
- **Doctor** reviews appointment requests and updates their status.
- **Amazon SNS** sends email notifications for appointment confirmations or rejections.
- **Response** is returned to the frontend and displayed to the user.



Name	Security group rule ID	Port range	Protocol	Source
-	sgr-002f4026b4ae89251	22	TCP	0.0.0.0/0
-	sgr-0408f11b2d577a2c2	0	TCP	0.0.0.0/0
-	sgr-01687da3694d74bed	80	TCP	0.0.0.0/0
-	sgr-036aa5dc5c918d9d5	3000	TCP	0.0.0.0/0

Figure 4.2: EC2 Instance Hosting the Backend

4.5 Database Design

The **Hospital Appointment Booking System** uses **MongoDB Atlas**, a cloud-based NoSQL database, to securely store and manage application data. The database is organized into multiple collections, each designed to store specific information required for the system. This structure enables efficient CRUD (Create, Read, Update, Delete) operations while maintaining data consistency and scalability.

- **Users Collection:** Stores patient and administrator details such as name, email, password, and role.
- **Doctors Collection:** Maintains doctor information including specialization, experience, consultation fee, and availability.
- **Appointments Collection:** Stores appointment details such as patient ID, doctor ID, appointment date, time, reason for visit, and status.
- **Reports Collection (Optional):** Stores consultation reports and treatment details if generated.
- **MongoDB Atlas:** Provides secure, scalable, and cloud-based data storage with high availability and reliable performance.

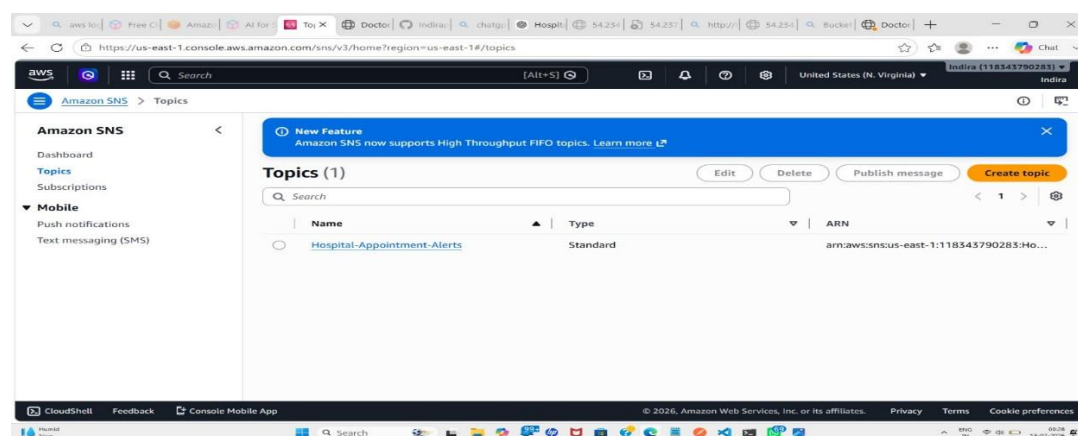


Figure : Simple Notification Service Dashboard

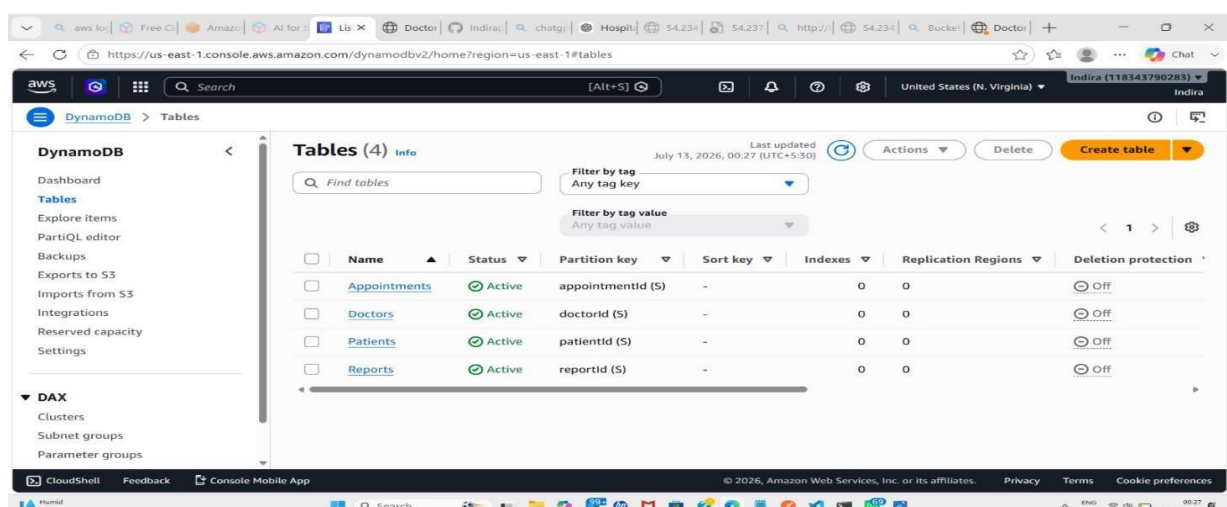


Figure: Amazon DynamoDB Tables

4.6 Security Design

The Hospital Appointment Booking System implements multiple security measures to protect user data, ensure secure access, and maintain the integrity of healthcare information. Authentication, authorization, and AWS security services are integrated to safeguard the application and its cloud resources.

- **User Authentication:** Secure login for patients, doctors, and administrators using email and password.
- **Role-Based Access Control:** Different permissions are assigned to patients, doctors, and administrators based on their roles.
- **Password Encryption:** User passwords are securely encrypted before being stored in the database.
- **AWS IAM:** Controls secure access to AWS resources by assigning appropriate permissions and policies.
- **Amazon SNS Security:** Sends appointment notifications only to registered email addresses.
- **Secure Database:** MongoDB Atlas provides encrypted cloud storage and secure database connectivity.
- **Input Validation:** Backend validates all user inputs to prevent invalid or malicious requests.

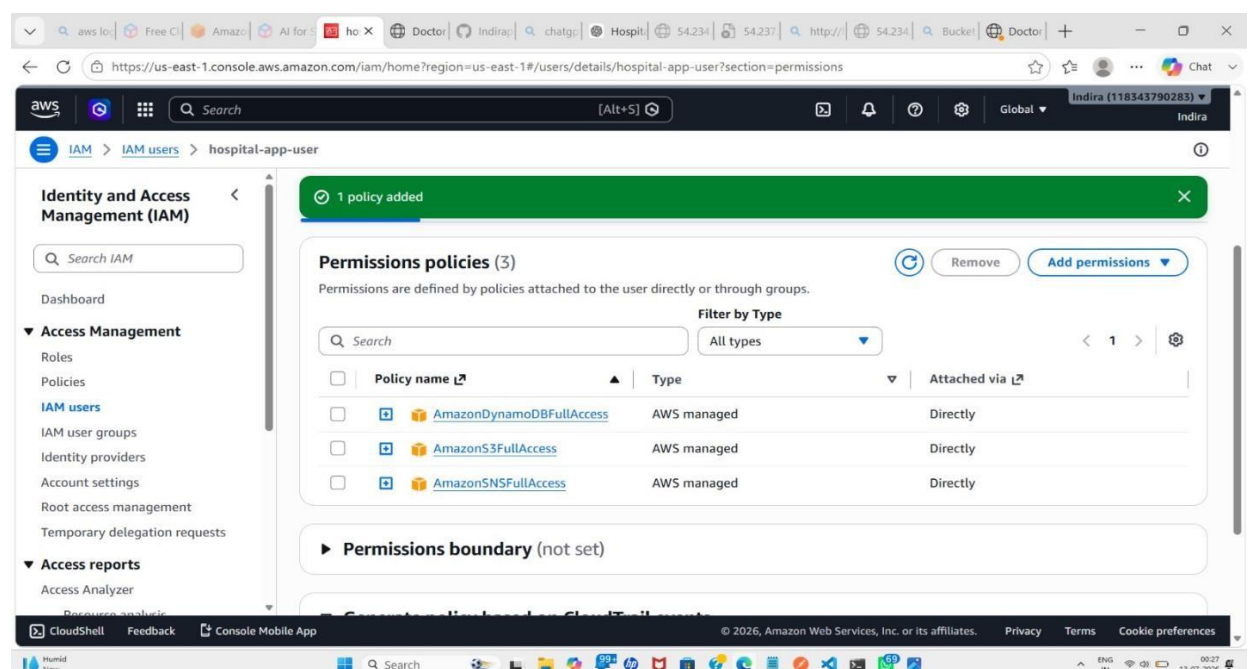


Figure 4.13: IAM Role for Secure Access

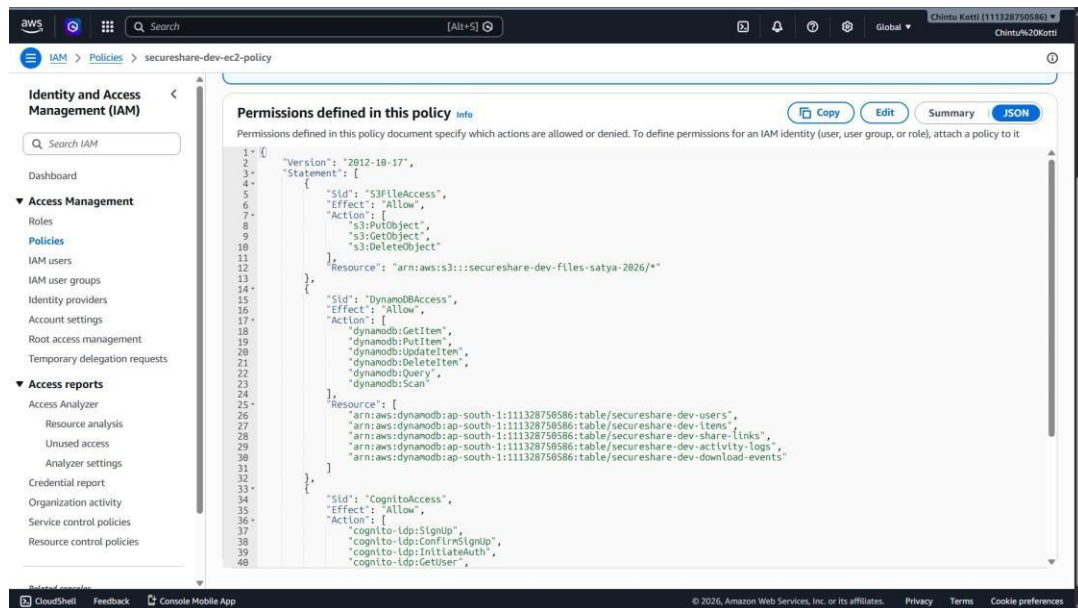


Figure 4.14: IAM Custom Policy Definition

Chapter 5

IMPLEMENTATION

5.1 Frontend Implementation

The frontend of the **Hospital Appointment Booking System** was developed using **React.js** to provide a responsive and user-friendly interface. It consists of separate pages for patients, doctors, and administrators, enabling role-based access to system functionalities. Patients can register, log in, search for doctors, and book appointments, while doctors and administrators can manage appointments and user information through dedicated dashboards. **React Router DOM** is used for page navigation, and **Axios** facilitates communication with the backend APIs. The interface is built using **HTML5, CSS3, and JavaScript**, ensuring a responsive design and smooth user experience across different devices.

5.2 Backend Implementation

The backend of the **Hospital Appointment Booking System** was developed using **Node.js** and **Express.js** to handle application logic and RESTful API requests. It manages user authentication, appointment booking, doctor management, and administrator operations. The backend communicates with **MongoDB Atlas** to store and retrieve patient, doctor, and appointment data securely. It also integrates **Amazon SNS** to send appointment notifications and **AWS IAM** to ensure secure access to cloud resources. The modular architecture, consisting of routes, controllers, models, and middleware, improves code organization, scalability, and maintainability while ensuring reliable application performance.

5.3 Authentication Implementation

The **Hospital Appointment Booking System** implements a secure authentication mechanism to ensure that only authorized users can access the application. Users, including patients, doctors, and administrators, must register and log in using valid credentials. The backend verifies user information before granting access to role-specific dashboards. Passwords are securely encrypted before being stored in the database, enhancing data security. Role-based access control ensures that each user can only access features relevant to their role. This authentication process protects sensitive healthcare information and provides a secure environment for managing appointments and user data.

5.4 Appointment Booking

The appointment booking module enables patients to schedule appointments with doctors through a simple and user-friendly interface. After logging into the system, patients can browse available doctors, select a preferred doctor, choose the appointment date and time, and submit the booking request. The backend validates the request and stores the appointment details securely in **MongoDB Atlas**. The booked appointment is then displayed on the doctor's dashboard for approval or rejection. Once the appointment status is updated, the system automatically sends an email notification to the patient using **Amazon SNS**, ensuring timely communication and efficient appointment management.

5.5 Admin Module

The **Admin Module** provides centralized control over the Hospital Appointment Booking System, enabling efficient management of hospital operations. Administrators can manage patient and doctor records, monitor appointments, and maintain overall system functionality through a dedicated dashboard. The module allows administrators to add, update, or remove doctor information and oversee appointment schedules to ensure smooth workflow. It also provides access to system statistics, helping administrators monitor healthcare services effectively. By centralizing administrative tasks, the module improves operational efficiency, reduces manual effort, and ensures accurate management of hospital resources.

5.6 Notification

The **Notification Module** is implemented using **Amazon Simple Notification Service (Amazon SNS)** to provide automated email notifications for appointment updates. When a doctor approves or rejects an appointment request, the backend triggers Amazon SNS to send a notification to the patient's registered email address. This ensures that patients receive timely updates without manually checking the application. The notification system improves communication between patients and healthcare providers, reduces administrative effort, and enhances the overall user experience. The integration of Amazon SNS provides a reliable, scalable, and secure cloud-based notification service.

5.7 Deployment

The **Hospital Appointment Booking System** was successfully deployed using **Amazon Web Services (AWS)** to provide a secure, scalable, and reliable cloud environment. The frontend application was hosted on **Amazon S3** using Static Website Hosting, allowing users to access the application through a web browser. The backend communicates with **MongoDB Atlas** to manage patient, doctor, and appointment data, while **Amazon SNS** handles automated email notifications. **AWS IAM** is used to manage secure access and permissions for AWS resources. This deployment ensures high availability, improved performance, and seamless access to the application from different locations.

Chapter 6

RESULTS

6.1 Final Output

The **Hospital Appointment Booking System** was successfully developed and deployed as a cloud-based web application. The completed system provides a user-friendly platform where patients can register, log in, search for doctors, and book appointments online. Doctors can review and manage appointment requests, while administrators can efficiently monitor doctors, patients, and appointment records through a centralized dashboard. The application integrates AWS cloud services to provide secure data management, frontend hosting, automated email notifications, and reliable cloud-based operation, ensuring a seamless healthcare appointment management experience.

6.2 Functional Results

The following functionalities were successfully implemented and verified during system execution:

- **User Authentication:** Patient, Doctor, and Administrator registration and secure login.
- **Appointment Management:** Online appointment booking, viewing, approval, rejection, and status updates.
- **Patient Module:** Doctor search, appointment history, and profile management.
- **Doctor Module:** Appointment request management and consultation scheduling.
- **Administrator Module:** Doctor management, patient management, and appointment monitoring.
- **Notification System:** Automated appointment status notifications using Amazon SNS.
- **Cloud Database:** Secure storage and retrieval of application data using MongoDB Atlas.

6.3 AWS Integration Results

The application successfully integrates the following AWS cloud services to provide a secure and scalable healthcare management platform:

- **Amazon S3** – Static website hosting for the frontend application.
- **Amazon SNS** – Automated email notifications for appointment updates.

- **AWS IAM** – Secure access control and permission management.
- **Cloud-Based Deployment** – Reliable application hosting with secure integration of AWS services.
- **MongoDB Atlas** – Cloud database for storing patients, doctors, and appointment records.

6.4 Deployment Results

The Hospital Appointment Booking System was successfully deployed using **Amazon Web Services (AWS)**. The frontend application was hosted on **Amazon S3** using Static Website Hosting, enabling users to access the system through a web browser. The backend was integrated with **MongoDB Atlas** for secure cloud-based data storage, while **Amazon SNS** was configured to send automated appointment notifications. AWS IAM was used to securely manage access permissions. The deployed application demonstrated reliable performance, secure communication, and seamless integration between the frontend, backend, database, and AWS services.

6.5 Testing Results

Comprehensive testing confirmed the successful operation of the system:

1. Patients, doctors, and administrators can register and log in securely.
2. Patients can search for doctors and book appointments successfully.
3. Appointment details are correctly stored and retrieved from MongoDB Atlas.
4. Doctors can approve or reject appointment requests through their dashboard.
5. Administrators can manage doctors, patients, and appointments efficiently.
6. Appointment notifications are successfully sent using Amazon SNS.
7. The frontend communicates correctly with backend APIs.
8. AWS services function reliably, ensuring secure and scalable application performance.

6.6 Output Screenshots

The following screenshots demonstrate the functional UI of the deployed Hospital Appointment booking application.

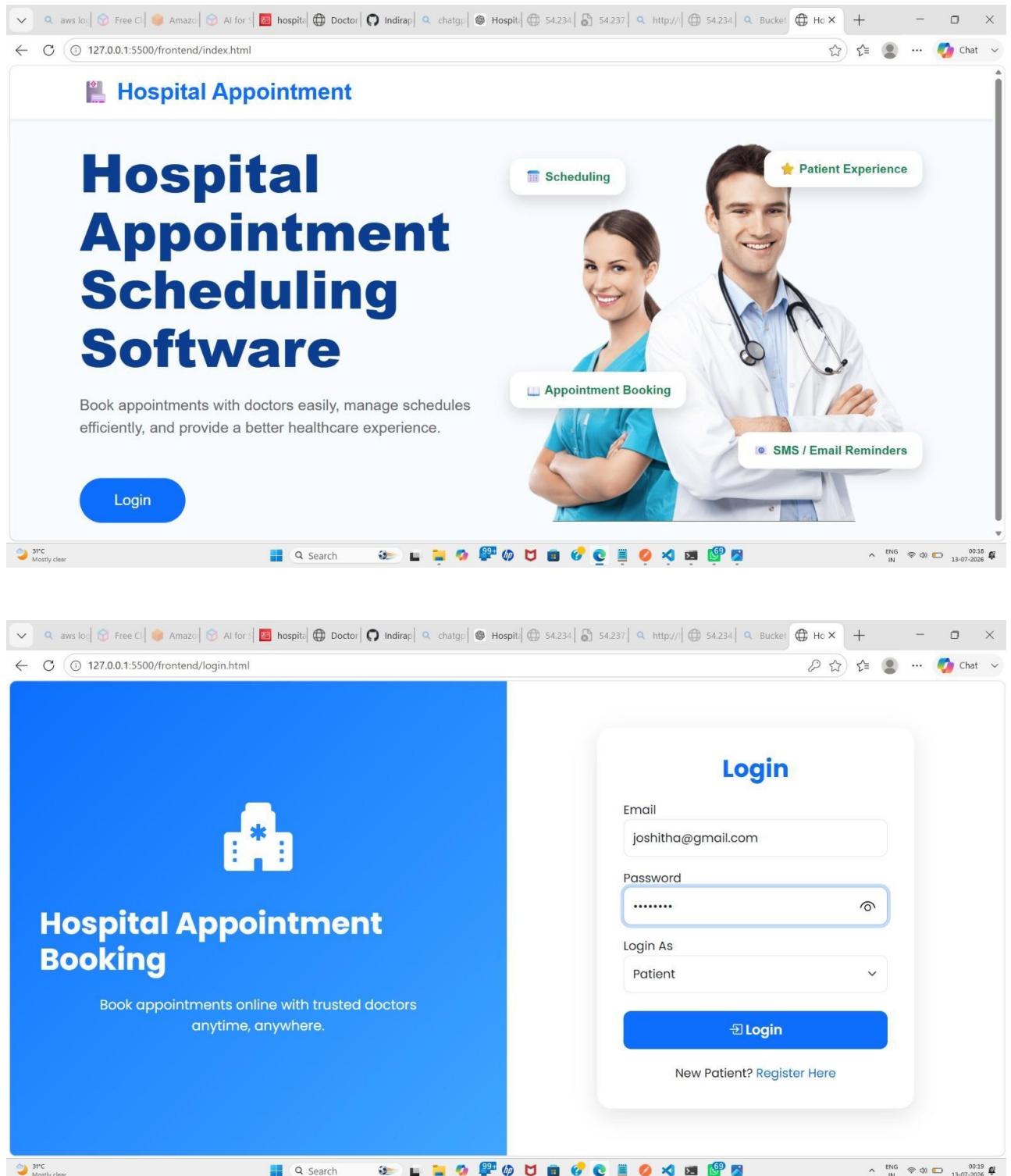


Figure 6.1: Login Page

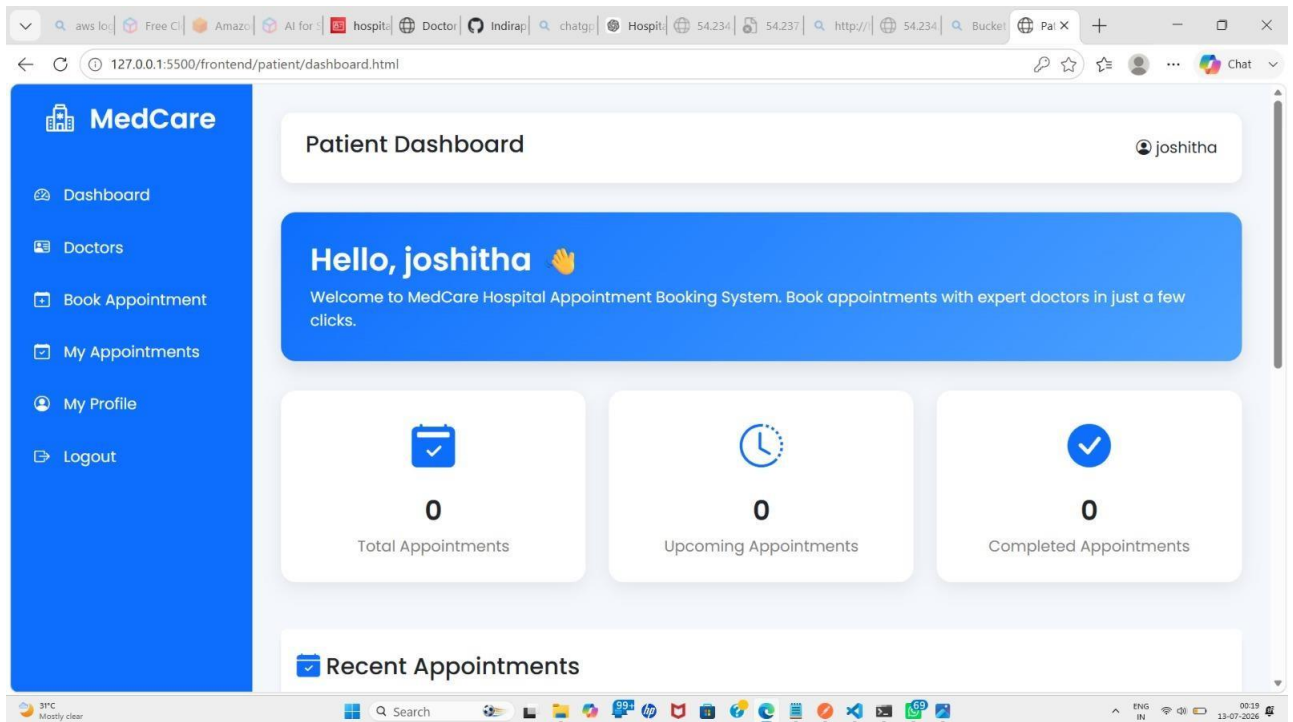


Figure 6.2: Patient Dashboard

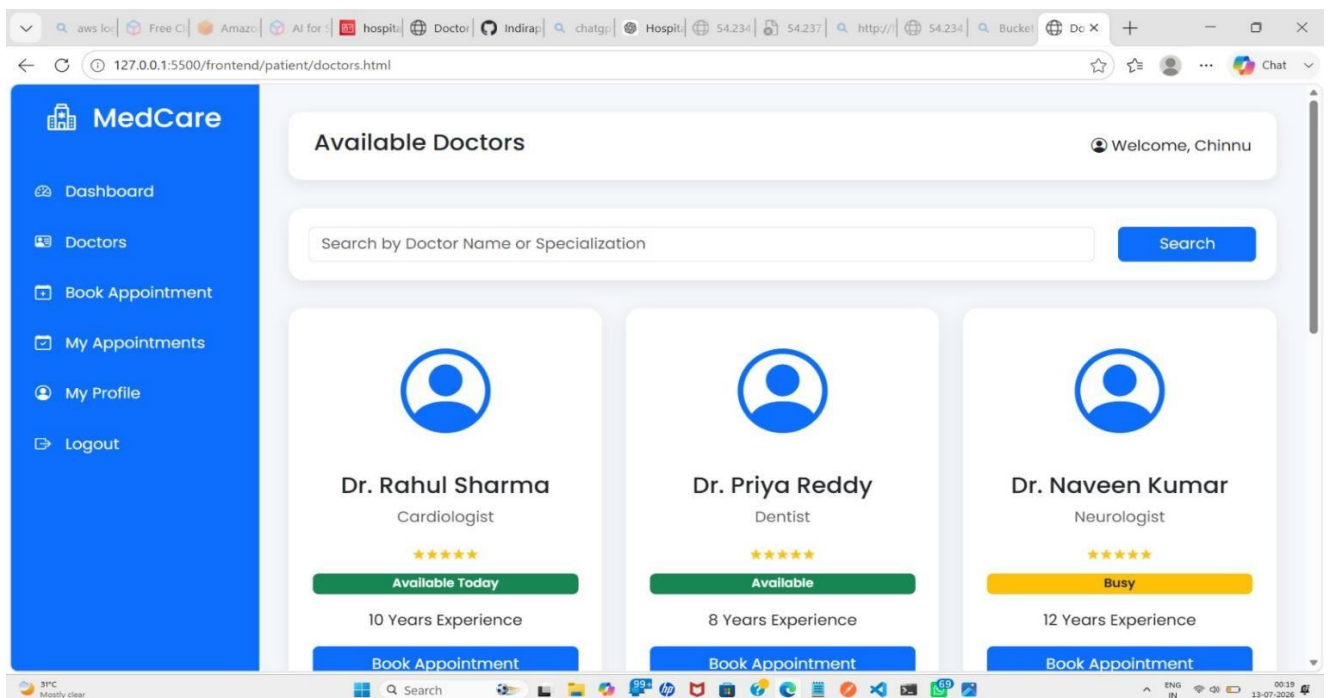


Figure 6.3: Available Doctors

MedCare

- Dashboard
- Doctors
- Book Appointment
- My Appointments
- My Profile
- Logout

Appointment Form

Patient Name
Chinnu

Email
chinnu@gmail.com

Select Doctor
Select Doctor

Department
Select Department

Appointment Date
dd-mm-yyyy

Appointment Time
--:--

Reason for Visit
Select Reason

Describe Your Symptoms
Briefly describe your health problem...

Additional Notes (Optional)
Any additional information...

Figure 6.4: Book Appointment Page

MedCare

- Dashboard
- Appointments
- Profile
- Logout

Doctor Appointment Dashboard

Dr. Rahul Sharma

28 Total Appointments

08 Pending

20 Completed

Search Patient: All Status: Search

Patient	Date	Time	Reason	Status	Action
Chinnu	15 Jul 2026	10:00 AM	Fever	Pending	View Accept Reject
Ravi Kumar	15 Jul 2026	11:30 AM	Heart Checkup	Confirmed	View Complete
Priya	16 Jul 2026	12:00 PM	Dental Pain	Completed	Details

Figure 6.5: Doctor appointment dashboard

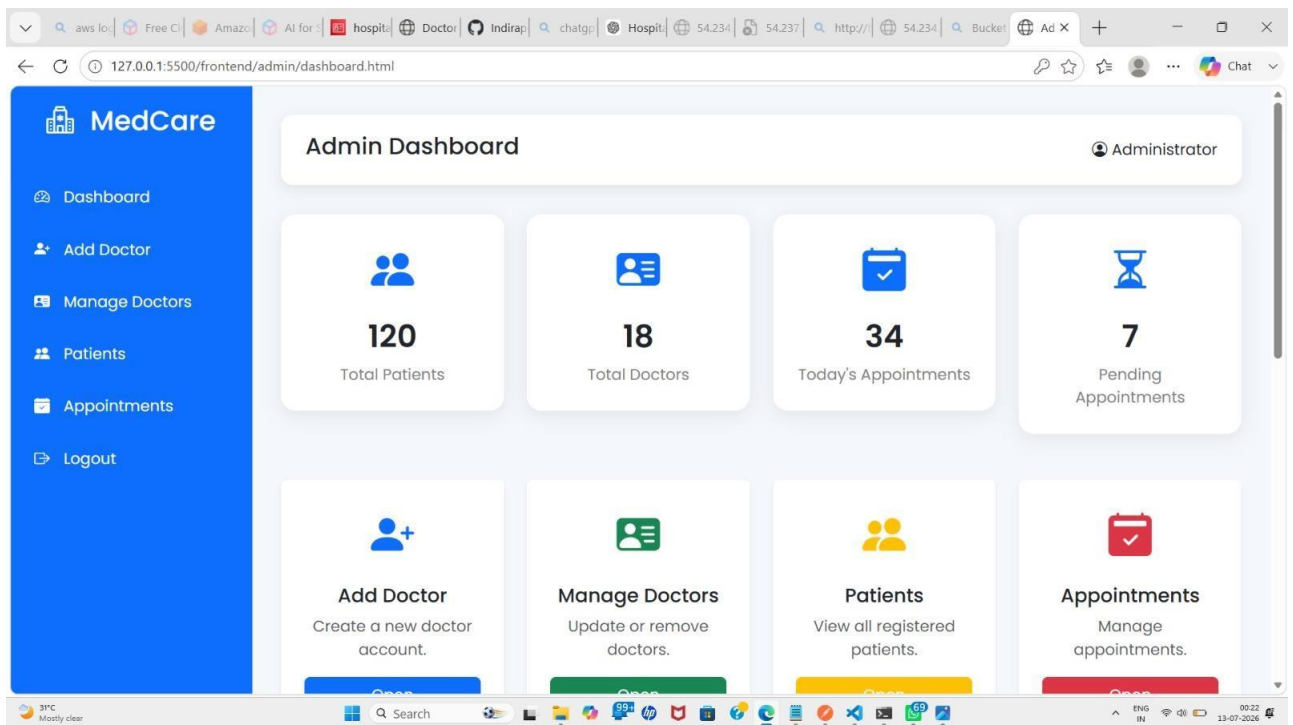


Figure 6.6: Admin dashboard

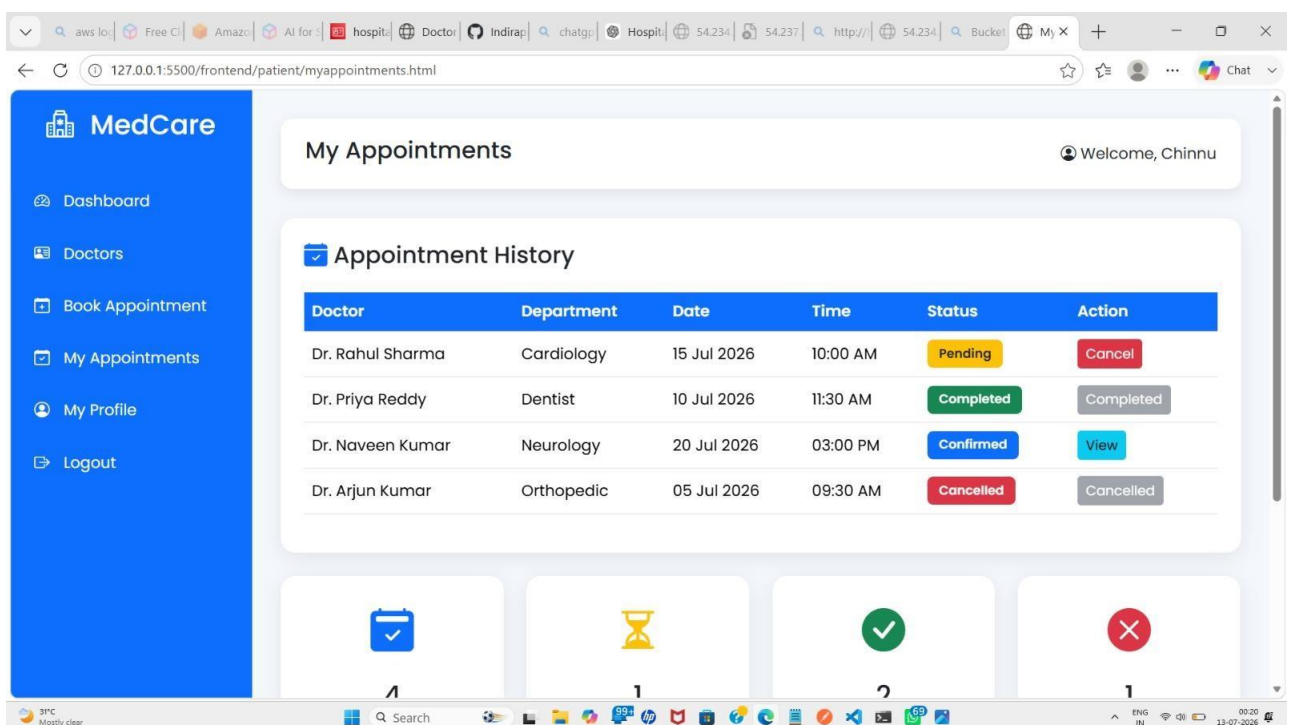


Figure 6.7: My Appointments.

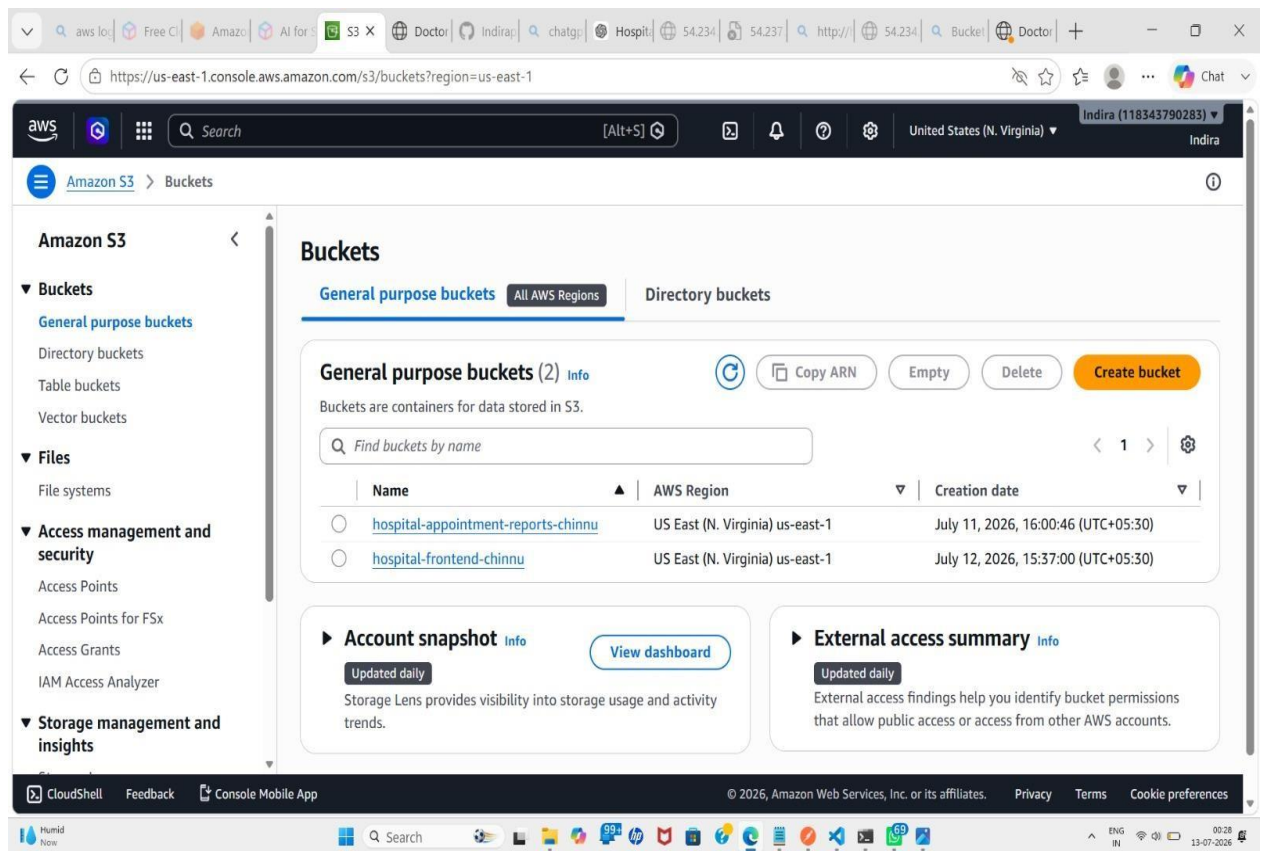


Figure 6.8: Amazon s3 bucket

Chapter 7

CONCLUSION

The Hospital Appointment Booking System successfully demonstrates the application of cloud computing technologies to improve healthcare appointment management. The system provides a secure, efficient, and user-friendly platform that enables patients to book appointments online, doctors to manage appointment requests, and administrators to oversee hospital operations through a centralized dashboard. By integrating React.js, Node.js, Express.js, MongoDB Atlas, and Amazon Web Services (AWS), the application delivers a scalable and reliable solution for modern healthcare institutions.

The implementation of **Amazon S3** for frontend hosting, **Amazon SNS** for automated email notifications, and **AWS IAM** for secure access management enhances the overall performance, security, and availability of the system. The project successfully reduces manual effort, minimizes appointment scheduling conflicts, improves communication between patients and healthcare providers, and ensures efficient management of hospital records.

Overall, the project achieved its objectives by developing a cloud-based appointment booking system that is secure, scalable, and easy to use. With future enhancements such as online payment integration, telemedicine, AI-based doctor recommendations, and mobile application support, the system has the potential to evolve into a comprehensive healthcare management platform suitable for real-world deployment.